GEOMETRIC POINT CRITERION FOR SET-THEORETIC PRE-EQUIVALENCE RELATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids-quotients.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The two details-omitted spots each reuse an argument displayed in full earlier in the same proof; no new technique is hidden.

Editorial note. The label lemma-set-theoretic-pre-equivalence-geometric names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter Definition definition-set-theoretic-equivalence, included as a context span below: j is a set-theoretic pre-equivalence relation when for every algebraically closed field k over B the relation sim-R on U(k), given by u sim-R u-prime iff some field extension K over k admits a K-valued point r of R with s(r) = u and t(r) = u-prime, is an equivalence relation; j is a set-theoretic equivalence relation when it is moreover universally injective
- The three induced maps of Equation equation-list: the composition map j-comp from R composed with itself over U, the diagonal map j-diag from U, and the flip map j-flip from R, all landing in U times over B U
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-large-enough: surjectivity statements on topological spaces can be witnessed on k-valued points as soon as the cardinality of the algebraically closed field k exceeds the bound lambda associated to R
- Morphisms of Spaces, Lemmas spaces-morphisms-lemma-permanence-finite-type and spaces-morphisms-lemma-base-change-finite-type: local finite type is preserved under composition and under base change
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-locally-finite-type-surjective-geometric-points: a nonempty algebraic space locally of finite type over an algebraically closed field has a k-valued point

Source reference. Original equation 048S: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 813–813. Prerequisite pointer invoked at 1097; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03XH: sources/stacks-chow/tags.txt, source lines 5148–5148. Prerequisite pointer invoked at 1214; it names a standard fact of a

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sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0488: sources/stacks-chow/tags.txt, source lines 5535–5535. Prerequisite pointer invoked at 1179; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0487: sources/stacks-chow/tags.txt, source lines 5534–5534. Prerequisite pointer invoked at 1217; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0462: sources/stacks-chow/tags.txt, source lines 5457–5457. Prerequisite pointer invoked at 1213; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Definition 0.1. Let S be a scheme, and let B be an algebraic space over S. Let $j : R \to U \times_B U$ be a pre-relation over B.

(1) We say j is a *set-theoretic pre-equivalence relation* if for all algebraically closed fields k over B the relation $\sim_R$ on $U(k)$ defined by

$$\overline{u} \sim_R \overline{u}' \Leftrightarrow \begin{array}{c} \exists \text{ field extension } K/k,\ \exists\ r \in R(K), \\ s(r) = \overline{u},\ t(r) = \overline{u}' \end{array}$$

is an equivalence relation.

(2) We say j is a *set-theoretic equivalence relation* if j is universally injective and a set-theoretic pre-equivalence relation.

Lemma 0.2. *In the situation of Definition 0.1. The following are equivalent:*

(1) *The morphism j is a set-theoretic pre-equivalence relation.*
(2) *The subset $j(|R|) \subset |U \times_B U|$ contains the image of $|j'|$ for any of the morphisms j' as in Equation (048S).*
(3) *For every algebraically closed field k over B of sufficiently large cardinality the subset $j(R(k)) \subset U(k) \times U(k)$ is an equivalence relation.*

If s, t are locally of finite type these are also equivalent to

(4) *For every algebraically closed field k over B the subset $j(R(k)) \subset U(k) \times U(k)$ is an equivalence relation.*

Proof. Assume (2). Let k be an algebraically closed field over B. We are going to show that $\sim_R$ is an equivalence relation. Suppose that $\overline{u}_i : \mathrm{Spec}(k) \to U$, $i = 1, 2$ are k-valued points of U. Suppose that $(\overline{u}_1, \overline{u}_2)$ is the image of a K-valued point $r \in R(K)$. Consider the solid commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K') & \dashrightarrow & \mathrm{Spec}(k) & \longleftarrow & \mathrm{Spec}(K) \\ \downarrow & & \downarrow{\scriptstyle (\overline{u}_2, \overline{u}_1)} & & \downarrow \\ R & \xrightarrow{j} & U \times_B U & \xleftarrow{j_{flip}} & R \end{array}$$

We also denote $r \in |R|$ the image of r. By assumption the image of $|j_{flip}|$ is contained in the image of $|j|$, in other words there exists a $r' \in |R|$ such that $|j|(r') = |j_{flip}|(r)$. But note that $(\overline{u}_2, \overline{u}_1)$ is in the equivalence class that defines $|j|(r')$ (by the commutativity of the solid part of the diagram). This means there exists a field extension K'/k and a morphism $r' : \mathrm{Spec}(K) \to R$ (abusively denoted r' as well) with $j \circ r' = (\overline{u}_2, \overline{u}_1) \circ i$ where $i : \mathrm{Spec}(K') \to \mathrm{Spec}(K)$ is the obvious map. In other words the dotted part of the diagram commutes. This proves that

$\sim_R$ is a symmetric relation on $U(k)$. In the similar way, using that the image of $|j_{diag}|$ is contained in the image of $|j|$ we see that $\sim_R$ is reflexive (details omitted).

To show that $\sim_R$ is transitive assume given $\overline{u}_i : \mathrm{Spec}(k) \to U$, $i = 1, 2, 3$ and field extensions K_i/k and points $r_i : \mathrm{Spec}(K_i) \to R$, $i = 1, 2$ such that $j(r_1) = (\overline{u}_1, \overline{u}_2)$ and $j(r_1) = (\overline{u}_2, \overline{u}_3)$. Then we may choose a commutative diagram of fields

$$\begin{array}{ccc} K & \longleftarrow & K_2 \\ \uparrow & & \uparrow \\ K_1 & \longleftarrow & k \end{array}$$

and we may think of $r_1, r_2 \in R(K)$. We consider the commutative solid diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K') & \dashrightarrow & \mathrm{Spec}(k) & \longleftarrow & \mathrm{Spec}(K) \\ \downarrow & & \downarrow{\scriptstyle (\overline{u}_1, \overline{u}_3)} & & \downarrow{\scriptstyle (r_1, r_2)} \\ R & \xrightarrow{j} & U \times_B U & \xleftarrow{j_{comp}} & R \times_{s,U,t} R \end{array}$$

By exactly the same reasoning as in the first part of the proof, but this time using that $|j_{comp}|((r_1, r_2))$ is in the image of $|j|$, we conclude that a field K' and dotted arrows exist making the diagram commute. This proves that $\sim_R$ is transitive and concludes the proof that (2) implies (1).

Assume (1) and let k be an algebraically closed field over B whose cardinality is larger than $\lambda(R)$, see Morphisms of Spaces, Lemma 0488. Suppose that $\overline{u} \sim_R \overline{u}'$ with $\overline{u}, \overline{u}' \in U(k)$. By assumption there exists a point in $|R|$ mapping to $(\overline{u}, \overline{u}') \in |U \times_B U|$. Hence by Morphisms of Spaces, Lemma 0488 we conclude there exists an $\overline{r} \in R(k)$ with $j(\overline{r}) = (\overline{u}, \overline{u}')$. In this way we see that (1) implies (3).

Assume (3). Let us show that $\mathrm{Im}(|j_{comp}|) \subset \mathrm{Im}(|j|)$. Pick any point $c \in |R \times_{s,U,t} R|$. We may represent this by a morphism $\overline{c} : \mathrm{Spec}(k) \to R \times_{s,U,t} R$, with k over B having sufficiently large cardinality. By assumption we see that $j_{comp}(\overline{c}) \in U(k) \times U(k) = (U \times_B U)(k)$ is also the image $j(\overline{r})$ for some $\overline{r} \in R(k)$. Hence $j_{comp}(c) = j(r)$ in $|U \times_B U|$ as desired (with $r \in |R|$ the equivalence class of $\overline{r}$). The same argument shows also that $\mathrm{Im}(|j_{diag}|) \subset \mathrm{Im}(|j|)$ and $\mathrm{Im}(|j_{flip}|) \subset \mathrm{Im}(|j|)$ (details omitted). In this way we see that (3) implies (2). At this point we have shown that (1), (2) and (3) are all equivalent.

It is clear that (4) implies (3) (without any assumptions on s, t). To finish the proof of the lemma we show that (1) implies (4) if s, t are locally of finite type. Namely, let k be an algebraically closed field over B. Suppose that $\overline{u} \sim_R \overline{u}'$ with $\overline{u}, \overline{u}' \in U(k)$. By assumption the algebraic space $Z = R \times_{j, U \times_B U, (\overline{u}, \overline{u}')} \mathrm{Spec}(k)$ is nonempty. On the other hand, since $j = (t, s)$ is locally of finite type the morphism $Z \to \mathrm{Spec}(k)$ is locally of finite type as well (use Morphisms of Spaces, Lemmas 0462 and 03XH). Hence Z has a k point by Morphisms of Spaces, Lemma 0487 and we conclude that $(\overline{u}, \overline{u}') \in j(R(k))$ as desired. This finishes the proof of the lemma. $\square$